

Project Presentation of Industry Oriented Hands on Experience (22CS422)

On

HealthHub (Smart Healthcare Management Platform)

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HealthHub is a web-based healthcare management platform designed to improve healthcare accessibility.

Traditional healthcare systems face problems like:

- Long waiting times
- Difficult appointment scheduling
- Limited doctor-patient communication
- Lack of emergency healthcare coordination

HealthHub provides:

- Online doctor appointment booking
- Real-time chat and video consultation
- Emergency blood request and donation system
- Nearby hospital find by map integration
- Centralized healthcare management

Technology Stack

- **Frontend:** React.js, TypeScript and Tailwind CSS
- **Backend:** Node.js and Express.js
- **Database:** MongoDB Atlas
- **Authentication:** JWT
- **Real-Time Communication:** WebRTC, Socket.io
- **Map-Integration:** Leaflet
- **Development Tool:** Visual Studio Code

HealthHub is a smart healthcare platform that connects patients, doctors, and donors through a centralized web application.

Main functionalities include:

- Online appointment booking
- Doctor profile Management
- Real-time chat and video consultation
- Emergency blood request system
- Dashboard for managing healthcare activities
- Map feature for nearby hospitals
- Secure authentication and data handling

HealthHub provides a secure, scalable, and responsive healthcare solution.

Objectives

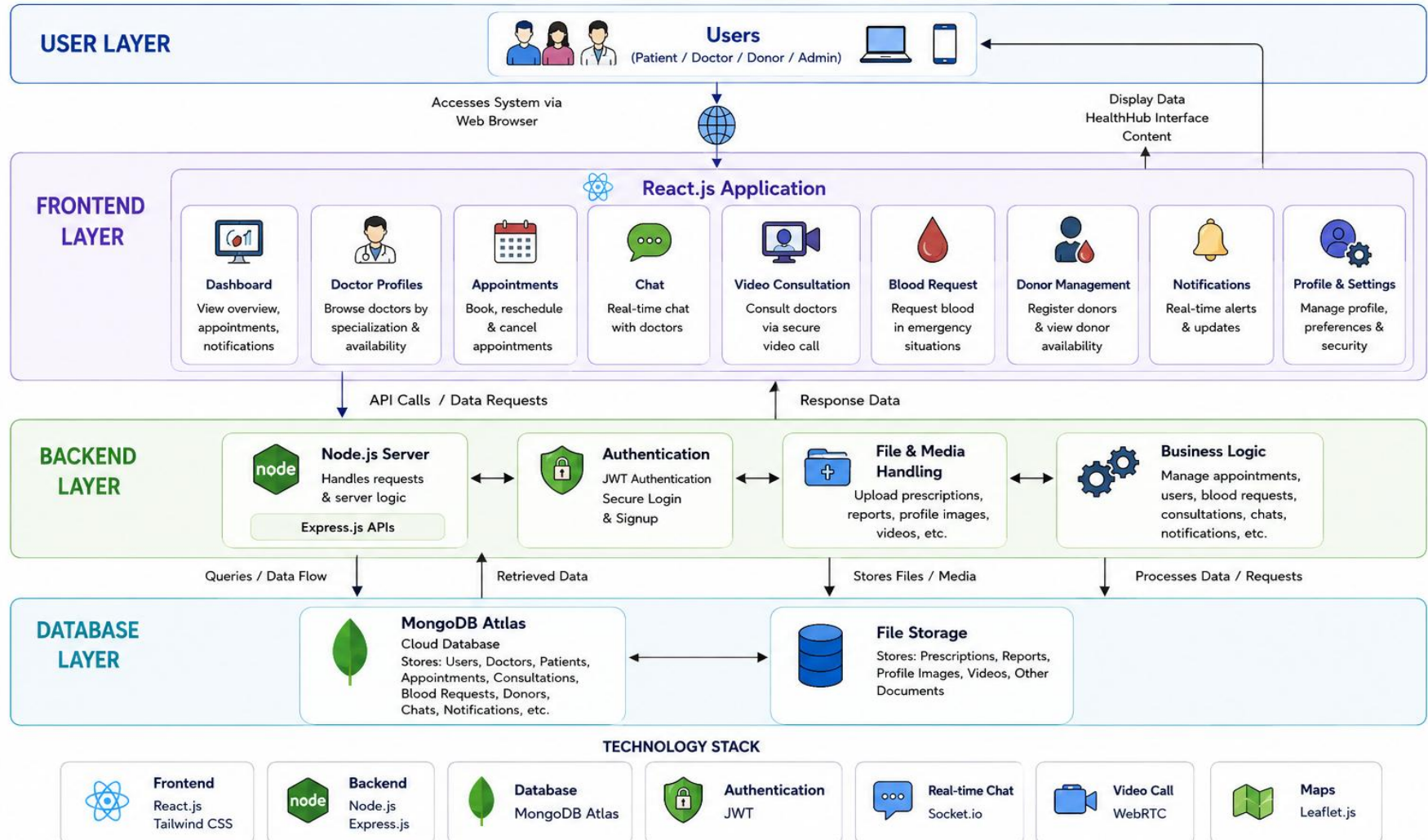
- To develop a smart healthcare management platform.
- To simplify online doctor appointment booking.
- To improve doctor-patient communication.
- To provide real-time chat and video consultation.
- To create an emergency blood request and donation system.
- To reduce manual healthcare management tasks
- To provide secure healthcare data management
- To improve accessibility of medical services
- To build a scalable and responsive healthcare platform

Key Features

- **User Authentication:** Secure login and signup using JWT
- **Appointment Management:** Book, update, and cancel appointments
- **Doctor Profiles:** View doctor specialization and availability
- **Real-Time Communication:** Live chat and video consultation
- **Emergency Blood Request:** Connect with blood donors quickly
- **Dashboards:** Patient and doctor activity dashboards
- **Notifications:** Real-time appointment and emergency updates
- **Map-Integration:** Search nearby hospitals within a area
- **Responsive Design:** Works smoothly on desktop and mobile devices

High-Level Design

HealthHub – Smart Healthcare Management Platform

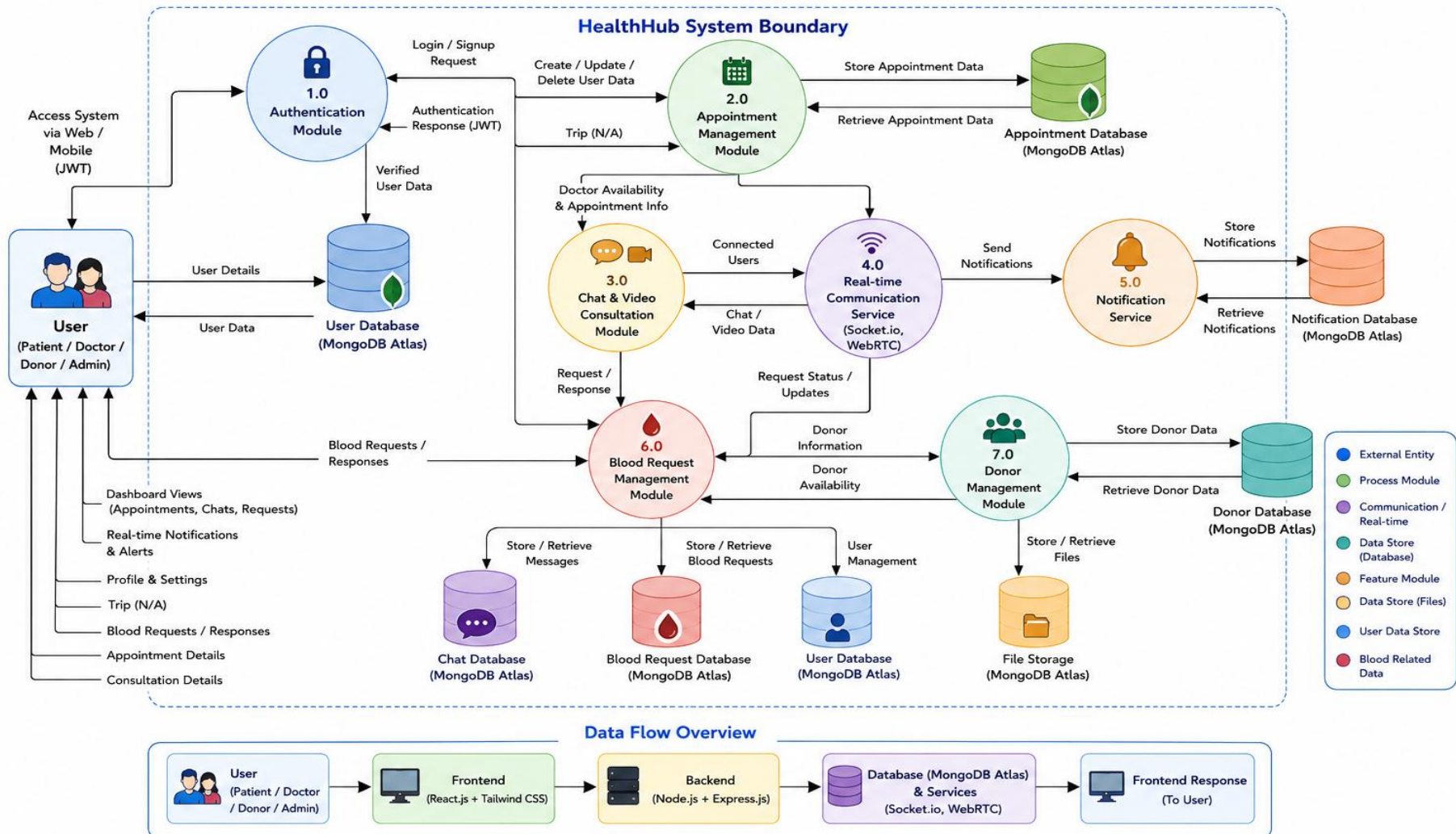


- The system follows a **3-tier architecture**: Frontend, Backend, and Database
- **Frontend (React.js + Tailwind CSS)** handles user interface and interactions
- **Backend (Node.js + Express.js)** processes requests and manages application logic
- Implements **JWT authentication** for secure login and access control
- **Real-Time Communication (Socket.io)** handles live chat and video calling between doctor and patient
- **Database (MongoDB Atlas)** stores users, doctors, blood-records, appointment and results
- Data flows from **User → Frontend → Backend → Database/Services → Frontend**.

Data Flow Diagram

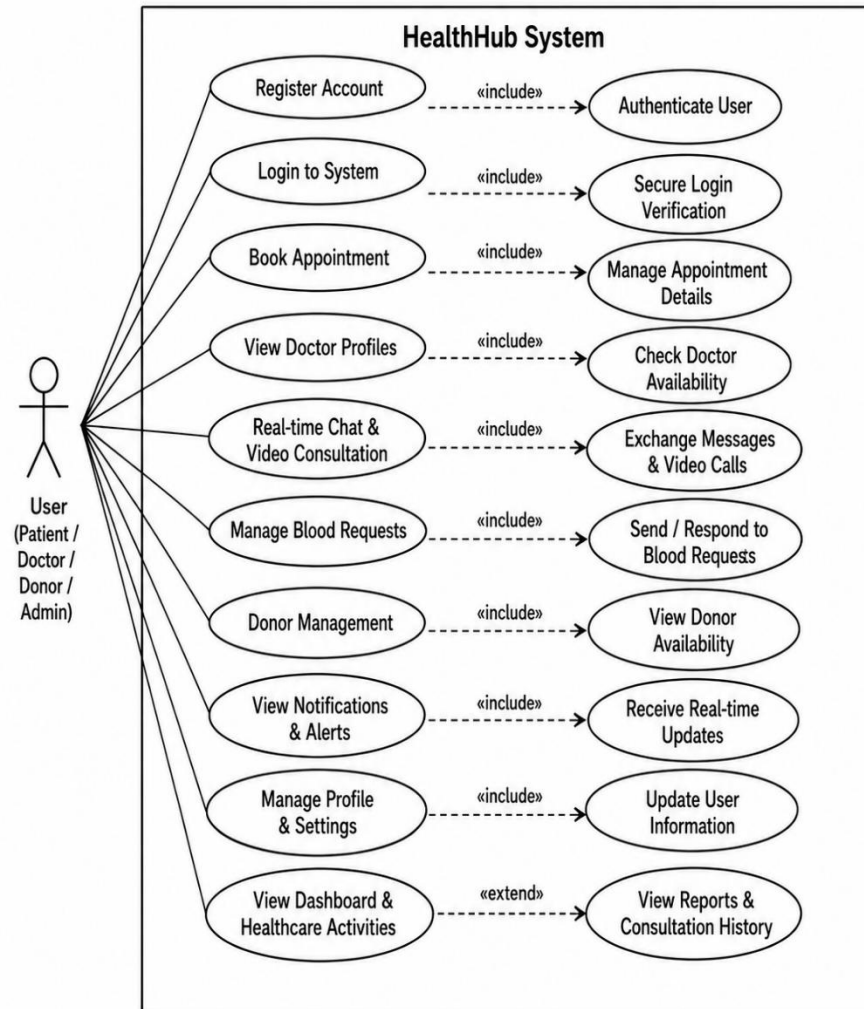
HealthHub – Smart Healthcare Management Platform

(DFD Level-1 Diagram)



Use Case Diagram

HealthHub – Smart Healthcare Management Platform
UML Use Case Diagram

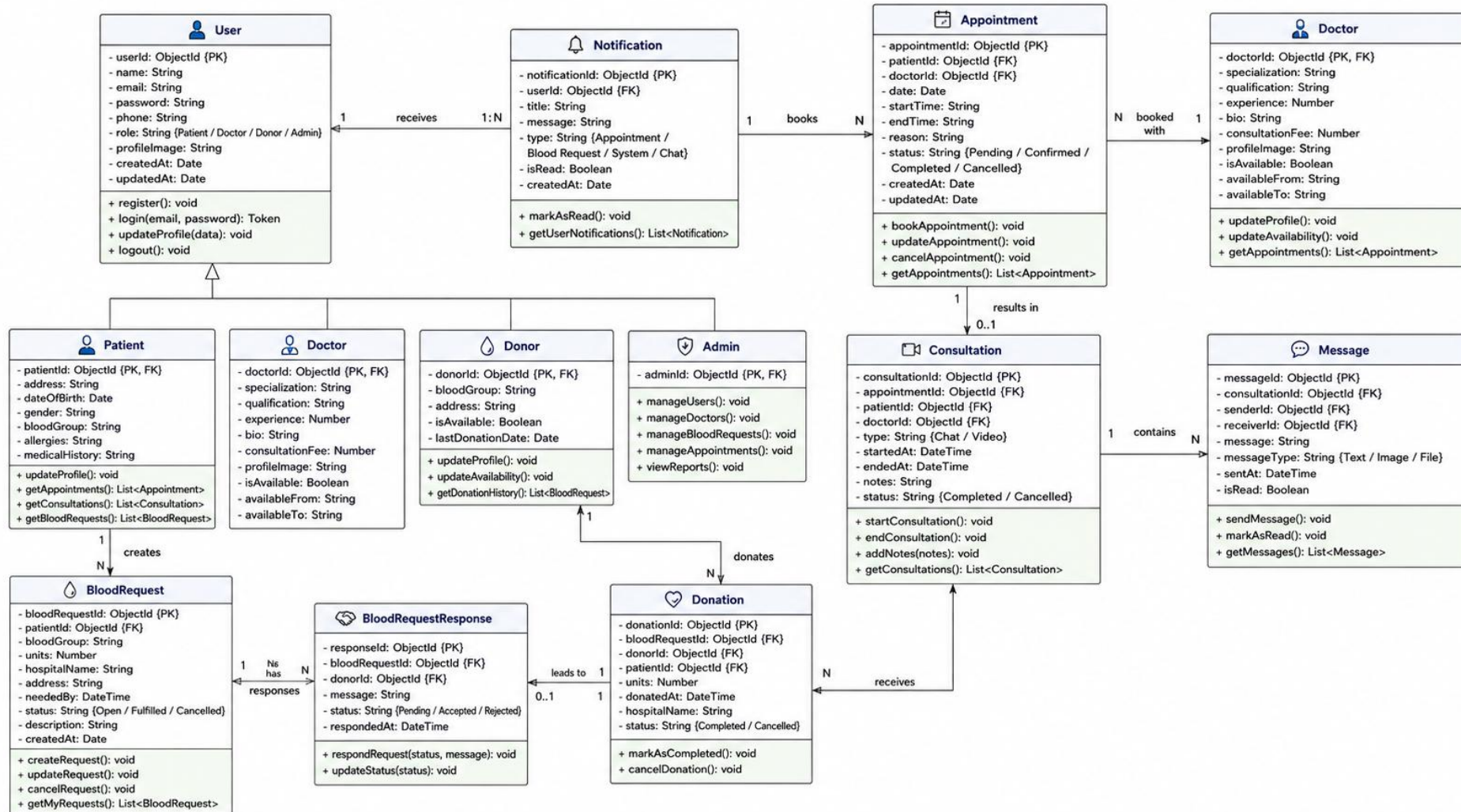


- Represents interaction between patients, doctors and the HealthHub system
- Shows all functionalities available in HealthHub.
- Users can register, login, and manage healthcare activities easily.
- Patients can book appointment and consult doctors through chat and video calls.
- Donors can respond to emergency blood requests quickly.
- Support real-time communication, notifications and healthcare management.
- Provides a complete smart healthcare management solution in one platform.

Class Diagram

HealthHub – Smart Healthcare Management Platform

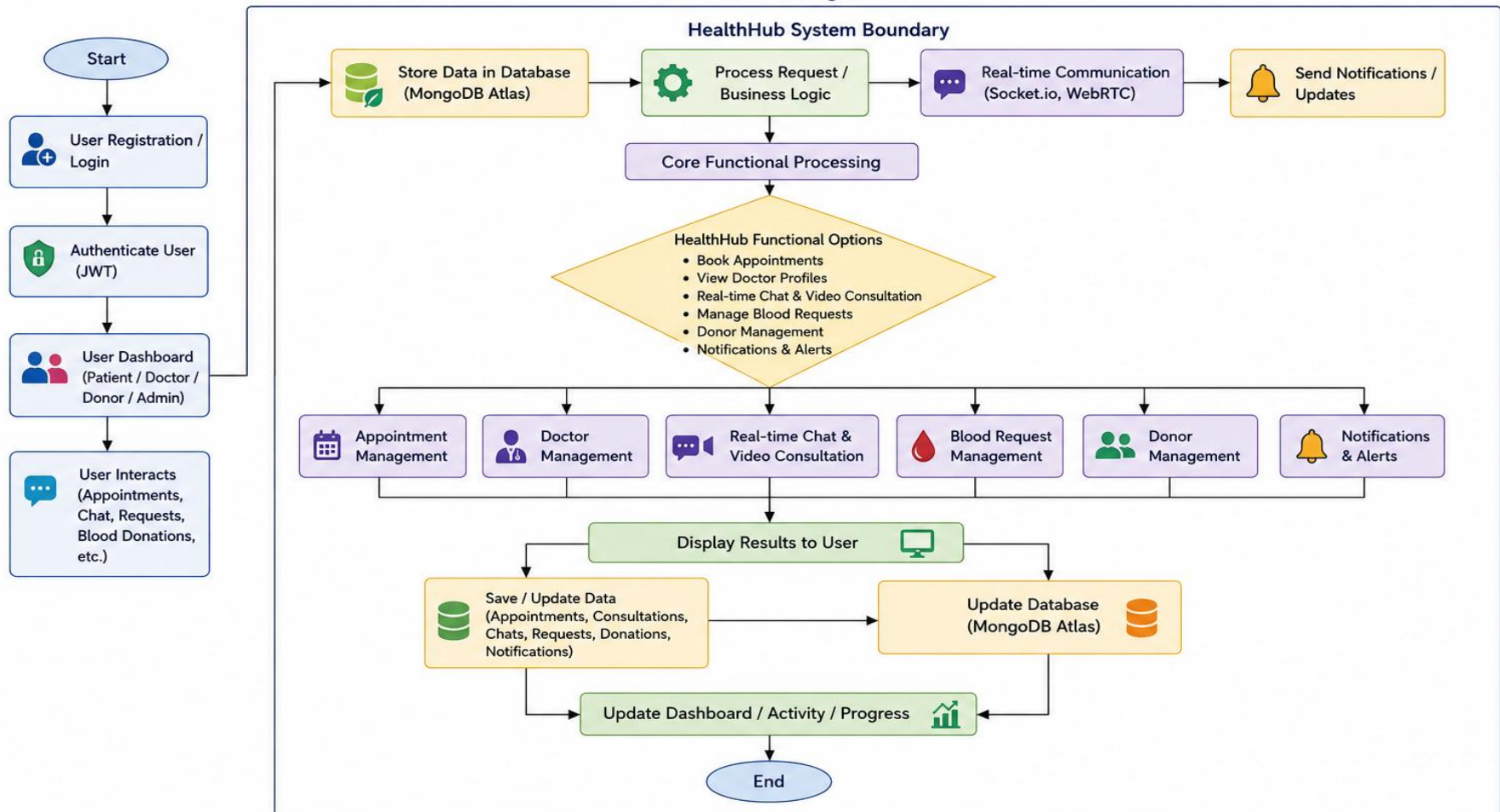
UML Class Diagram



Flow Chart

HealthHub – Smart Healthcare Management Platform

Flowchart Diagram



Testing & Results

Testing:

- Unit Testing for appointment, authentication, and communication modules.
- Integration Testing for frontend-backend interaction.
- Functional Testing for all healthcare workflows.
- UI Responsiveness Testing.
- Real-Time Chat and Video Testing.

Results:

- Appointment booking worked successfully.
- **Real-time** communication performed smoothly.
- **Blood request** system generated fast responses.
- **Dashboard** displayed healthcare activities correctly.
- Platform delivered secure and efficient healthcare services.

Limitations

- **Real-time chat** and notifications depend on stable internet connectivity.
- **Video consultation quality** may vary with network speed.
- **Emergency blood response** depends on donor availability
- **Offline support** is currently limited
- **Large scale** simultaneous users may affect performance
- **AI-based** healthcare recommendations are not yet implemented

Future Scope

- AI-based health recommendations
- Integration with wearable health devices
- Advance healthcare analytics dashboard
- Mobile application deployment
- Online medicine ordering system
- Multi-Language support
- Cloud Scalability improvements
- Ai chatbot for healthcare assistance

Thank You